Workshop No. 1 — Conceptual Design for a Domotic Circuit Simulator

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Requirements Documentation:

Functional requirements:

- The program must allow the user to create circuits from scratch.
- The program must simulate circuit behavior in real time.
- The user must be able to save and load circuit designs for future use.
- The system must provide error messages when invalid connections or missing components are detected.
- The simulator must enable users to connect components visually to define data and power flows.

Non-Functional requirements:

- The simulator must provide an user-friendly interface for the users.
- The simulation must process updates as quickly as possible.
- The codebase must be documented and structured to facilitate maintenance and future extensions.
- The system must provide tooltips or short explanations for each button and component.
- The system should include an example for new users.

User Stories:

Title	Component data	Priority	Estimate
User story	As a user, I want to view the data of the components used in the circuit.	High	1 month
Acceptance criteria	When I need to build the home automation circuit, it should provide me with the data of each component in the circuit.		

Title	Connect components	Priority	Estimate
User story	As a user, I want to connect multiple components to create a circuit.	High	1 month
Acceptance criteria	When I need to build a circuit, it should allow me to place and connect several components together.		

Title	Component catalog	Priority	Estimate
User story	As a user, I want to have access to a wide variety of different components to build various types of circuits I need.	High	1 month
Acceptance criteria	When using the application, the user should have a diverse list of components (bulbs, resistors, switches, sensors, etc.).		

Title	Pleasant and accessible interface	Priority	Estimate
User story	As a student using the simulator, I want to visualize the bulb’s energy consumption when it is turned on, in order to analyze its impact on the circuit.	Medium	2 months
Acceptance criteria	When accessing the application, the user should be able to easily navigate and use the program’s functions without confusion.		

Title	Modify component values	Priority	Estimate
User story	As a user, I want to modify the values of the components in the circuit.	High	1 month
Acceptance criteria	When running the simulation, the system should calculate and display the consumption in real time or as an accumulated value.		

Title	Add custom components	Priority	Estimate
User story	As a user, I want to add different components for specific tasks.	High	1 month
Acceptance criteria	When required, a new component can be added to the main catalog.		

Title	Energy consumption simulation	Priority	Estimate
User story	As a student using the simulator, I want to visualize the bulb’s energy consumption when it is turned on, in order to analyze the impact of using different devices on the electrical system.	High	1 month
Acceptance criteria	When required, a new component can be added to the main catalog.		

CRC CARDS:

Class	Sensor
Responsibility	**Collaborator**
Detect changes in the environment (movement, temperature, light). Send readings to a controller / processor. Activate something according to the measured values.	Controller Bulb Actuator

Class	Controller
Responsibility	**Collaborator**
Coordinate sensors and actuators. Process the system logic. Display the overall system status.	Sensor Actuator Interface

Class	Actuator
Responsibility	**Collaborator**
Perform physical actions. Receive commands from the controller. Confirm its status (on/off).	Controller Sensor

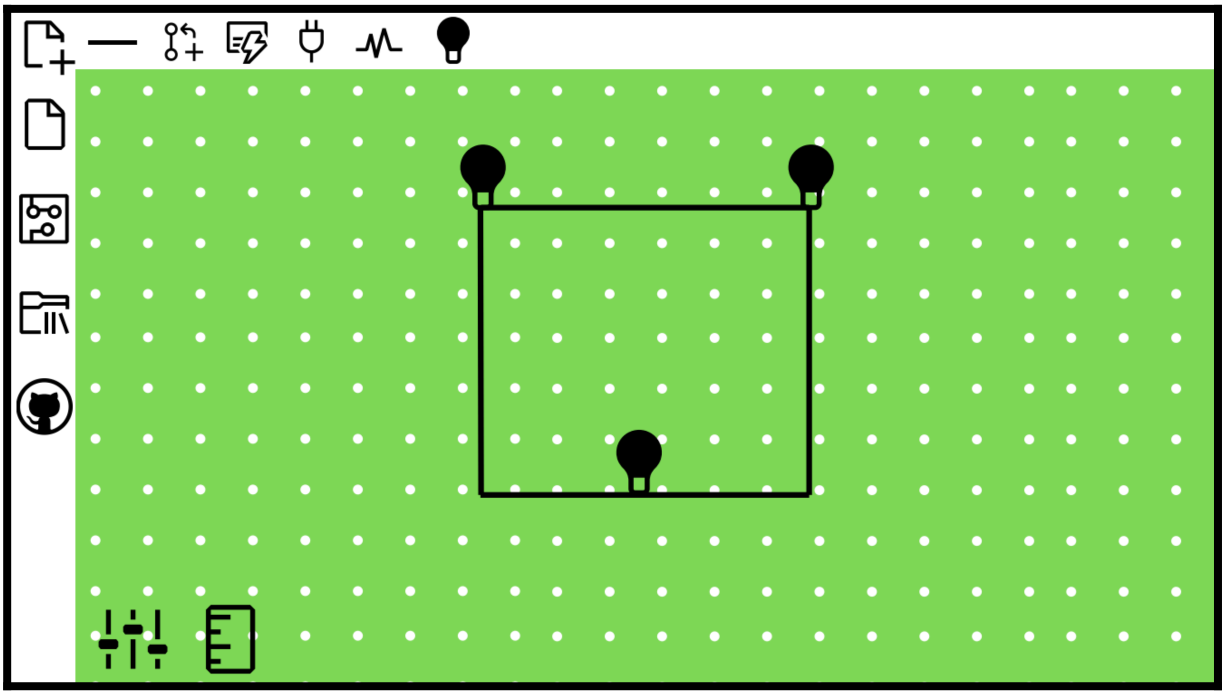
Class	Interfaz Usuario
Responsibility	**Collaborator**
Display the status of sensors and actuators. Allow the user to configure or control devices. Send commands to the controller.	Controller

Class	Simulador de Circuito
Responsibility	**Collaborator**
Create and manage the system instances (sensors, actuators, controller). Run the simulation step-by-step. Display the circuit's results as visuals or text.	Sensor Actuator Controller User Interface

Class	Bulb
Responsibility	**Collaborator**
Turn on or off upon receiving a signal. Display its current state (on/off). Show energy consumption data.	Switch Controller Sensor

Class	Switch
Responsibility	**Collaborator**
Change its state (on/off). Send a signal to the bulb and/or controller.	Controller Bulb

Mockups:



On the left side, you can create new projects, open existing ones, import components, and generate the preview and simulation of the current circuit.
At the top section, you can integrate different components as well as their respective connections.
In the green dotted area, you will find the design workspace, where two buttons allow you to measure and modify the parameters of the components.

References:

Our inspiration comes from circuit simulators such as **QUCS (Quite Universal Circuit Simulator)**.

Github:
https://github.com/danielclavijo23/CLASE-POO